

What is Virtualization?

Virtualization – the technology that powers the cloud.

Real-Life Problem

Imagine you're a **software developer** who has created a new application.

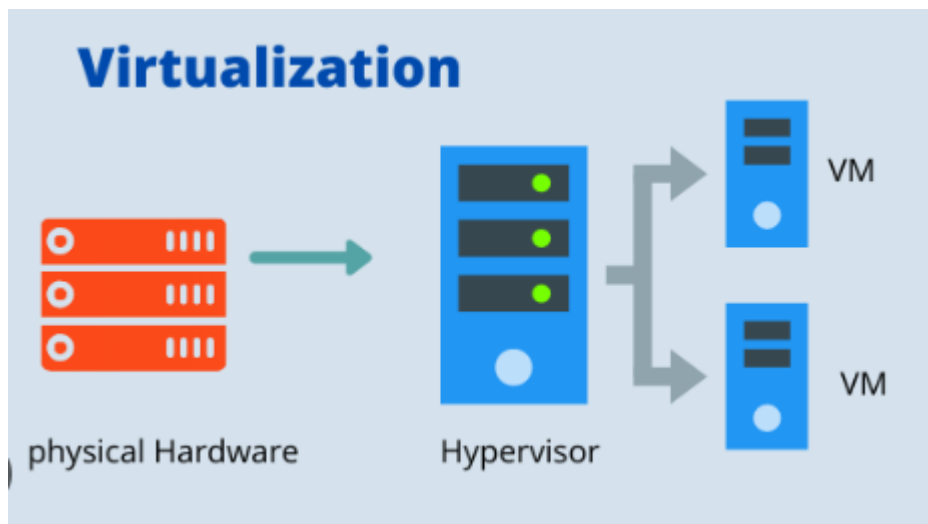
You want to test how your app works on **different operating systems**, like **Windows**, **Linux**, and **Ubuntu**.

Normally, this would require **separate machines** for each OS — which is **expensive and hard to manage**.

That's where **virtualization** saves the day!

What is Virtualization?

Virtualization is the process of running **multiple virtual machines (VMs)** on a **single physical machine** using the system's resources (CPU, RAM, storage, etc.).



Before virtualization, running multiple applications required **multiple physical servers**, which meant higher costs, more maintenance, and underutilized resources.

Each VM acts like a **separate computer** and can run its own **operating system**.

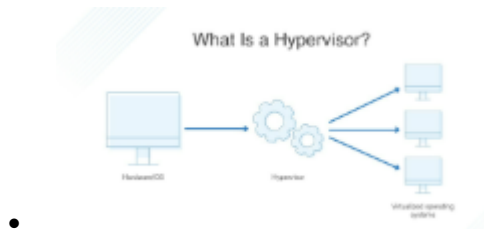
✓ This means:

- You can run Windows, Linux, and Ubuntu all on the same machine
- No need to buy 3 separate systems
- Saves time, money, and physical space
- **How Does Virtualization Work?**

At the center of virtualization is a powerful software called the **Hypervisor**.

🔧 Hypervisor:

A **hypervisor** is a software layer that sits between your **hardware** and the **virtual machines**. It manages and allocates resources (CPU, RAM, Disk) to each VM and allows them to run independently.



🧩 Types of Hypervisors

There are **2 types** of hypervisors:

1 Type 1: Bare Metal Hypervisor

- Runs **directly on physical hardware**
- Faster and used in **data centers**
- Example: **VMware ESXi, Microsoft Hyper-V, Xen**

2 Type 2: Hosted Hypervisor

- Runs **on top of an existing OS**
- Easy to use for **personal testing or learning**
- Example: **VirtualBox, VMware Workstation**

📖 Example in Simple Words

Let's say you have a **laptop** with good specs (8 GB RAM, i5 CPU). You want to test your software on:

- **Windows 10**
- **Linux Ubuntu**

Instead of buying a new system, you install **VirtualBox (Type 2 Hypervisor)** on your laptop.

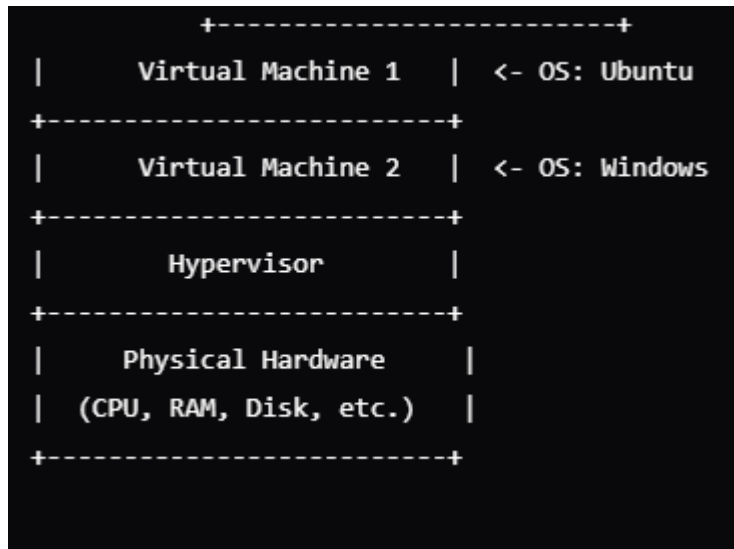
Inside VirtualBox, you create:

- One VM with **Windows 10**
- One VM with **Ubuntu**

Now both are running at the same time on your single laptop.

- That's the power of virtualization!

Diagram – How Virtualization Works



Why Virtualization is Important in Cloud (AWS)

AWS uses virtualization to offer services like **EC2 instances**, which are **virtual servers**.

You don't need to buy or manage physical servers – just **launch a VM**, install the OS, and run your app.

Benefits of Virtualization

- Run multiple OS on one system
- Save cost and energy
- Easy to test and deploy software
- Isolated environments for each VM
- Foundation for **cloud computing**